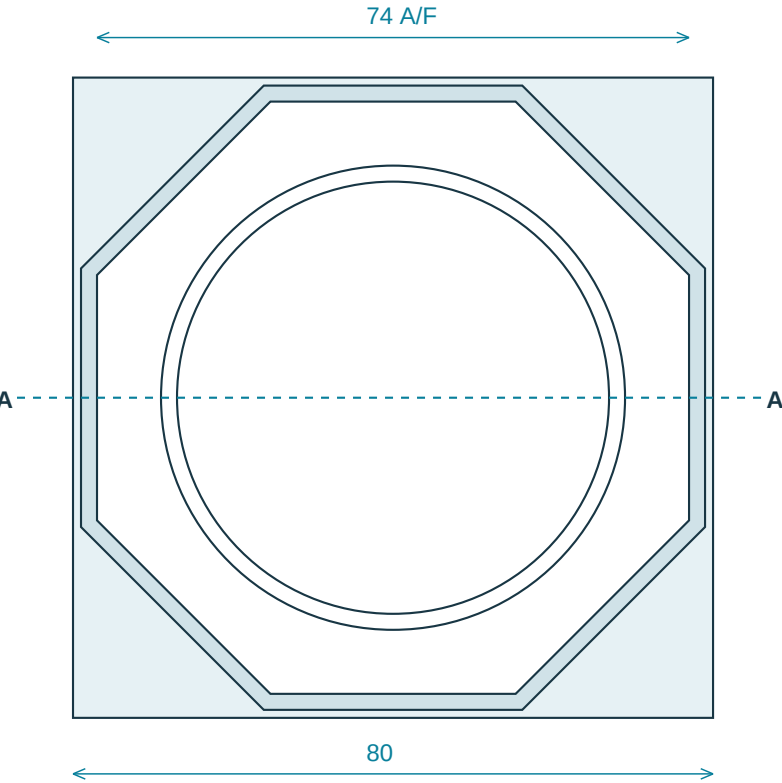


WINDOWED OCTAGONAL CAGE

UMC-11 / REV A AL6061 AMBITIOUS PROJECT-AUTHORED CHALLENGE

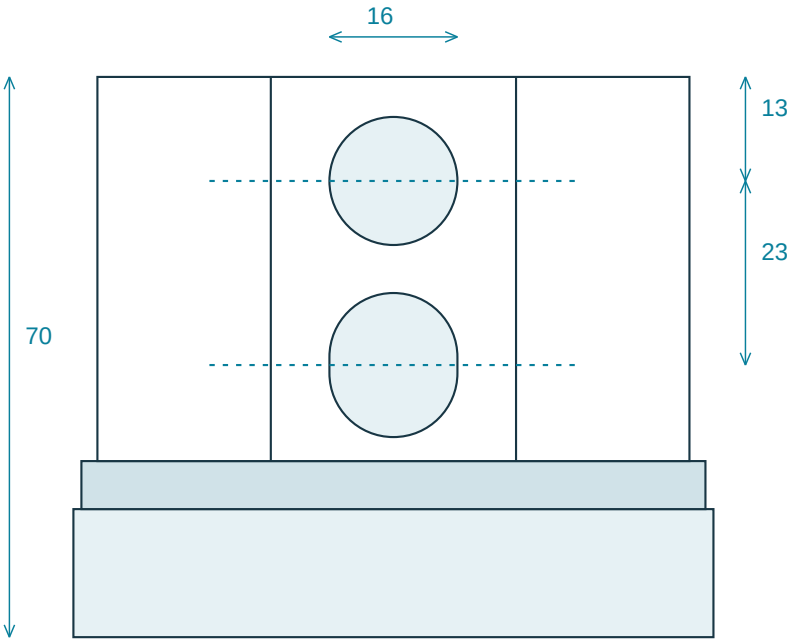
01 ORTHOGRAPHIC DEFINITION - ALL DIMENSIONS IN MM

TOP / LOOKING -Z



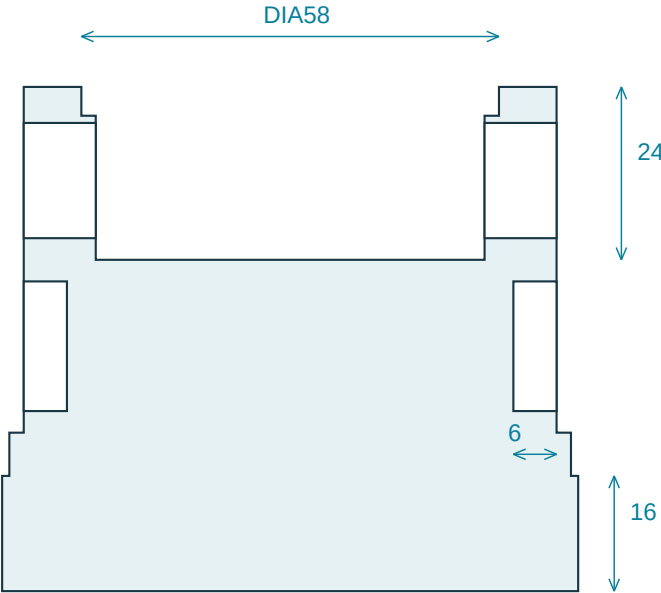
Main body 74 A/F, Z=-48..0.
Collar 78 A/F, Z=-54..-48.
Square base 80 x 80, Z=-70..-54.
Top: DIA58 x 4; DIA54 x 24 total depth.

TYPICAL FACET / LOOKING ALONG -N



8X upper: DIA16 through near wall only.
Centers U=0, Z=-13; normal removal 14.
8X lower: 16 wide x 18 high, R8.
Centers U=0, Z=-36; blind depth 6.

SECTION A-A / THROUGH X AXIS



Section cuts two opposing upper windows.
Top cavity floor Z=-24; no hidden chamber.
Upper window Z=-21..-5, opens into cavity.
Lower relief Z=-45..-27; floor N=31.

DATUMS / INTERPRETATION

Origin: stock top center (X,Y,Z)=(0,0,0). Stock bottom Z=-70.
All octagonal facet outward normals theta=0,45,90,135,180,225,270,315 degrees from +X toward +Y.
 $N=X \cos(\theta)+Y \sin(\theta)$; $U=-X \sin(\theta)+Y \cos(\theta)$. The typical facet drawing applies to all eight faces.
Upper windows intentionally meet the top cavity. Lower relief pockets are blind; their lower core remains solid.
General tolerance +/-0.127 mm; angular +/-0.2 deg. Sharp convex edges; no extra chamfers, fillets or features.

FIXTURE: preserve full 80 x 80 x 16 bottom flange. No base side/bottom machining or mounting-hole creation.
Machine: existing Haas UMC-750, indexed 3+2; T1 only, DIA12.7 flat, flute 25.4, stickout 33.02.
See sheet 2 for authoritative feature coordinates and analytical access limits.

WINDOWED OCTAGONAL CAGE

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02 FEATURE COORDINATES / ACCESS SCREEN

SOLID ENVELOPE

Keep base: X,Y=-40..40; Z=-70..-54. Add regular octagonal collar 78 A/F at Z=-54..-48.

Above collar: regular octagon 74 A/F at Z=-48..0. Octagon = intersection of eight $N \leq A/F/2$ halfspaces.

Collar has 2 mm radial step beyond main faces. No decorative radius or corner blend is implied.

OPEN TOP CAVITY

Concentric with origin: DIA58 circular counterbore from Z=0 to -4; DIA54 cavity from Z=0 to -24.

Both floors planar normal to Z. Resulting DIA54 floor is Z=-24; the annular shoulder is Z=-4.

No inner island, taper, undercut, thread or lower internal cavity.

8X UPPER THROUGH-WALL WINDOWS

One on every theta facet: circle $U^2 + (Z+13)^2 \leq 64$. Center U=0, Z=-13; DIA16.

Remove along N=23..37 (14mm normal depth from outer face). This deliberately intersects DIA54 cavity.

At U=8, cavity near wall $N = \sqrt{27^2 - 8^2} = 25.7876$; N=23 endpoint gives 2.7876 mm breakthrough margin.

Do not extend the window cut through the opposite cage wall. Circular window is visible at N37.

8X LOWER BLIND RELIEF WINDOWS

One on every theta facet: vertical capsule 16 wide, 18 high, R8; center U=0, Z=-36.

Semicircle centers U=0,Z=-35 and U=0,Z=-37. Straight sides $U = \pm 8$; total Z=-45..-27.

Remove along N=31..37 only (6mm normal depth); retain flat floor N=31. No internal island.

These do not connect into the cavity or into adjacent pockets; they are not a second hollow chamber.

RIBS / TOOL ACCESS - ANALYTICAL SCREEN, NOT VERIFICATION

DIA16 apertures exceed T1 DIA12.7; all internal planar profile radii R8 exceed tool R6.35.

Top depth 24 < flute 25.4 and stickout 33.02; cavity bottom leaves 9.02 mm axial holder standoff from top.

Facet width 30.652 mm; 16 mm window leaves 7.326 mm on each facet edge. Internal rib arc approximately 4.962 mm.

Upper window bottom -21 is 3 mm above cavity floor. Lower relief top -27 is 3 mm below that floor.

Vertical gap between window levels 6 mm; lower relief ends 3 mm above collar. Counterbore rim 8 mm.

Worst diagonal outside-stock removal 19.569 mm < flute 25.4. Upper body height 48 requires indexed access.

No blind depth beyond tool reach is specified. Workholding, engagement, holder/fixture clearances, stock conformity and actual manufacturability still require the agent and full Fusion verification.